

Jack Schedel

Developing modern and efficient software

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Education

Bachelor of Science - Dec 2024

Computer Science
University of Florida
Cum Laude

Coursework

- Digital Logic and Computer Systems
- Microprocessor Applications
- Real Time Operating Systems
- Programming Language Concepts
- Algorithm Abstraction and Design
- Advanced Systems Programming
- Human-Computer Interaction
- Bioinformatic Algorithms
- Distributed Operating Systems

Skills

- Go
- Typescript / Next.js / Astro
- Rust
- C / C++
- React / Tailwind CSS
- Drizzle ORM / Sequelize
- Neovim / Lua
- Python / Jupyter Notebooks
- Embedded STM32 / MSP430
- Postgres / MySQL
- AWS / Azure / Vercel
- C# / .NET
- Java / Kotlin

Work Experience

Software Engineer, Microsoft

Jan 2025 - Present

- Building Azure .NET services full-time as part of Microsoft Entra ID. Designing and shipping C# features for the Lifecycle Workflows product, partnering across teams to deliver reliable, scalable business-to-business functionality.
- Own features end-to-end, from design and implementation through testing, deployment, and ongoing support in production.

Strategy Lead, FSGP 2023 Champions, Solar Gators

Feb 2023 - Jul 2025

- Developed a real-time telemetry data analysis tool for our team's solar-powered racecar to run live regression calculations in the racetrack pit using TypeScript. Used the analysis tools to manage and determine team race strategy during the 2023 Formula Sun Grand Prix, in which our team won the overall competition.
- Created a race physics simulation CLI tool using Go, used in conjunction with a Python optimization solver using mystic to run a gradient descent optimization maximizing energy efficiency to find the optimal driving strategy. Used to generate an optimal velocity heatmap for each point in the 11-turn track.

Azure Software Engineering Intern, Microsoft

Summer 2024

- Wrote C# code for a business-to-business Azure .NET service as part of Microsoft Entra ID. Developed a feature for the Lifecycle Workflows team with full product and architectural ownership in a 12 week timespan.
- Designed an optimized implementation, cleanly built into the existing complex multiservices architecture. Received a hire-back offer from the same team, where I now work full-time.

Embedded Systems Intern, Oak Ridge National Laboratory

Summer 2023

- Wrote C and C++ code deployed to MSP430 chips on custom PCBs for a localized tracking system designed for use with containers of fissile nuclear material.
- Implemented ultra-wideband multilateration to determine container locations, calculating distance information between container devices and anchor devices around a target area with precisely placed anchor devices.

Projects

AutoCalibr, Generative Autoencoder for 3D Meshes

- Created and trained a custom variational autoencoder model using Keras to generate new meshes using principle component analysis in the latent space.
- Handled comprehensive preprocessing of raw mesh data including face tri conversion, volume/positioning normalization, and face subdivision padding.
- Generated thousands of different random variations of each object during normalization and padding to artificially expand the dataset and enhance the model's comprehension of the intrinsic properties of the mesh data.

KoalaClient, LLM Playground Interface

- Developed an advanced interface for the OpenAI API using TypeScript and Tailwind CSS to better integrate LLMs into my personal workflow.

EndGame2, UCI Chess Engine

- Developed a UCI-compatible chess engine written in Rust that uses minimax tree-traversal over all possible board states using a custom board-state evaluation function.
- Implemented alpha-beta pruning, zobrist position hashing, and capture-resolve quiescence searching to improve search times and engine performance.

RTOS Tetris, Custom RTOS for Tiva C Series

- Developed a custom RTOS from the ground up, to be used with Tiva C Series processors, and programmed a fully featured Tetris game from scratch in it.
- Optimized the thread context switching and minimized unnecessary redraws, resulting in a snappy and lag-free gameplay experience, despite only running on a 16MHz processor and very slow display.